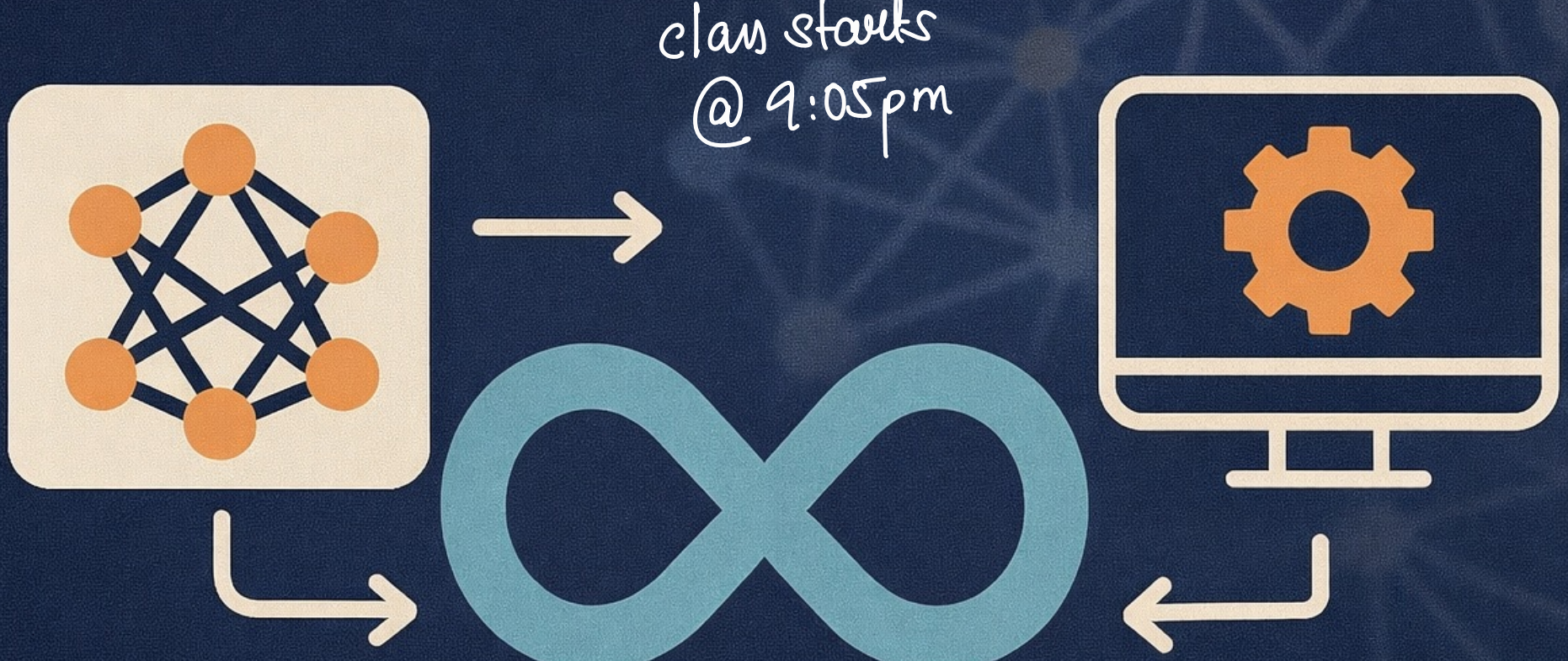
NEURAL NETWORK



Introduction

"Build & Break things"

Aniruddha Mukherjee

(Ani)

MLE ⟶ 8+ YOE

Micron Technology

2+ YOE

Scaler

Lecture 4: Tensorflow Keras

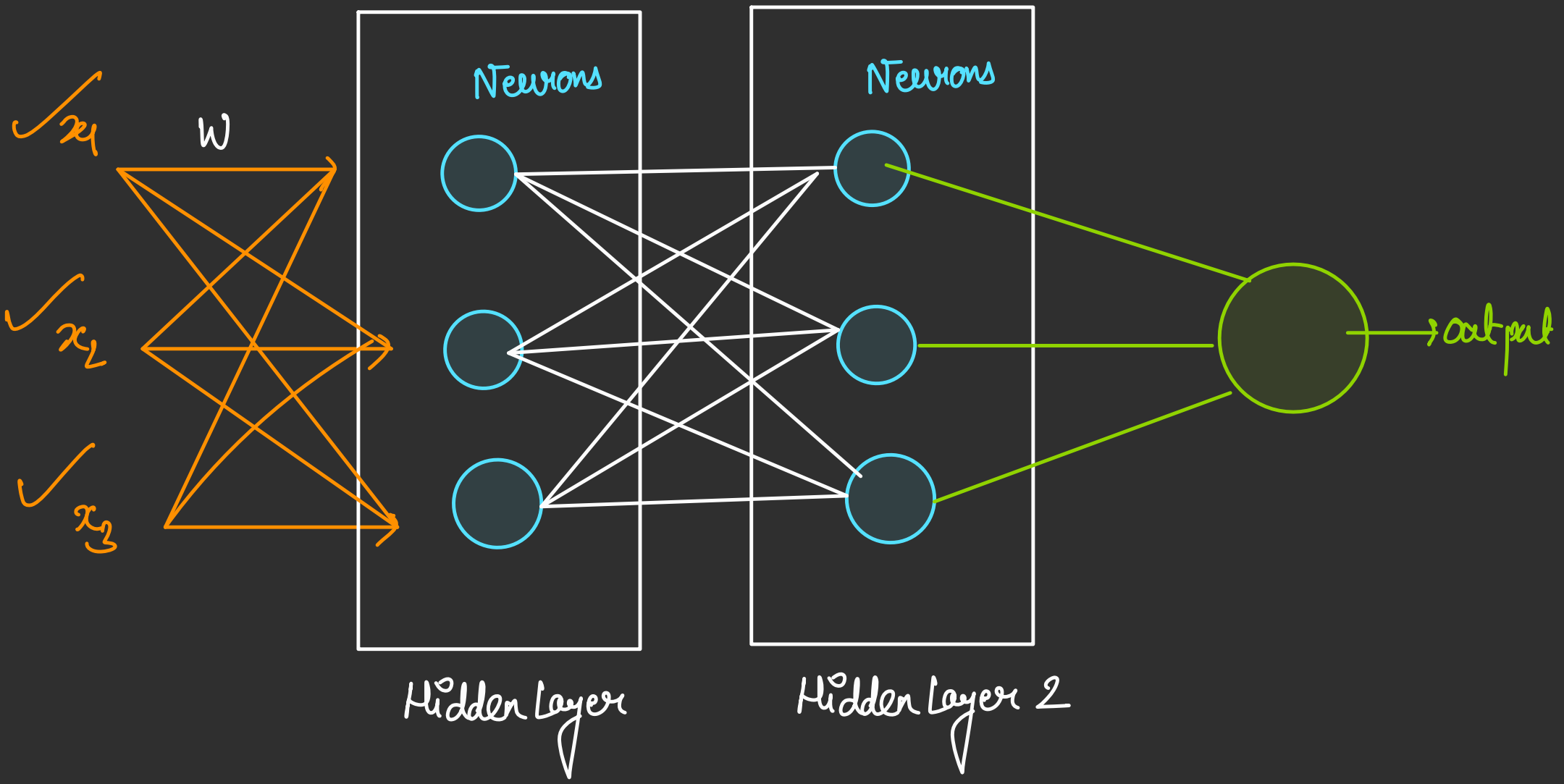
#Agenda

1) Basic Neural Network – Simple Encoding + dense Layer
2) Dropout
3) Batch Normalization

"CASE STUDY"

Question → Q/A

Neural Network (FC layers)



Case Study

Textual data (Reviews)

AI generated

Human written

classification (Binary)

- Logistic Reg → linear dec
- Random Forest → Non-

Why Neural Network?

$w_1x_1 + w_2x_2$

1) Smaller data → Classical ML

   Larger data → NN

2) Manual FE → Classical ML

   No Manual FE → NN

3) Solve the curse of dimentionity.

Image (200 × 200 × 3) ✓

Textual

Classical ML ?

(NN / Classical ML) → YES

Smaller data

NN

Class ML ✓

1) Tokenization

text => "I love pizza"

sentence

Breaking sentence into smaller unit → words

I - 5

love - 23

pizza - 11

[5 23 11]

Train

I love pizza

Test

Burgur

↳ Out of Vocabulary (OOV)

2) Padding & Truncation

I love pizza

3

Seq: [5 23 11]

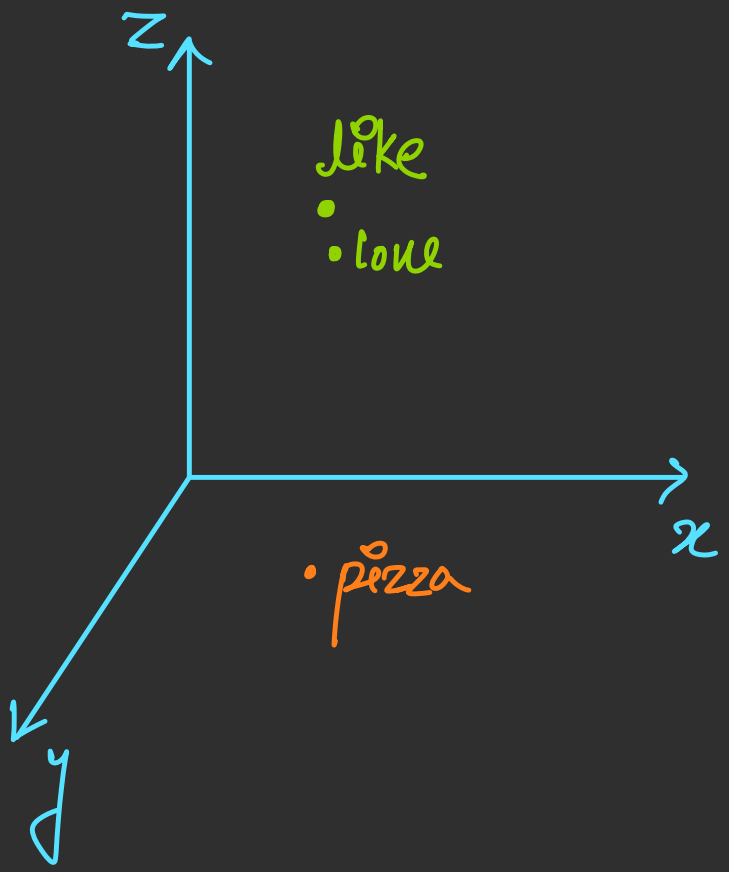
Seq: [5 23 11 0]

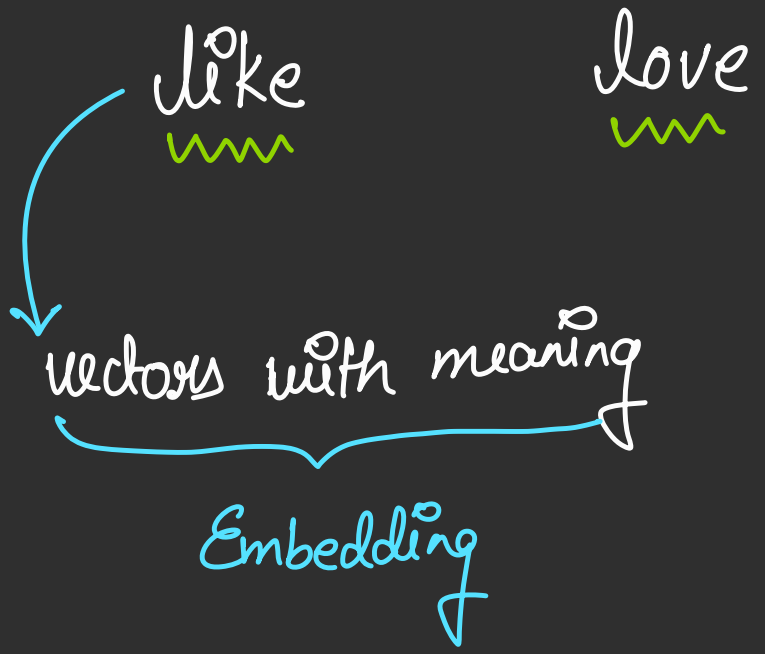
I hate golf games

4

Seq: [5 4 36 82]

3) Embeddings : giving meaning to numbers

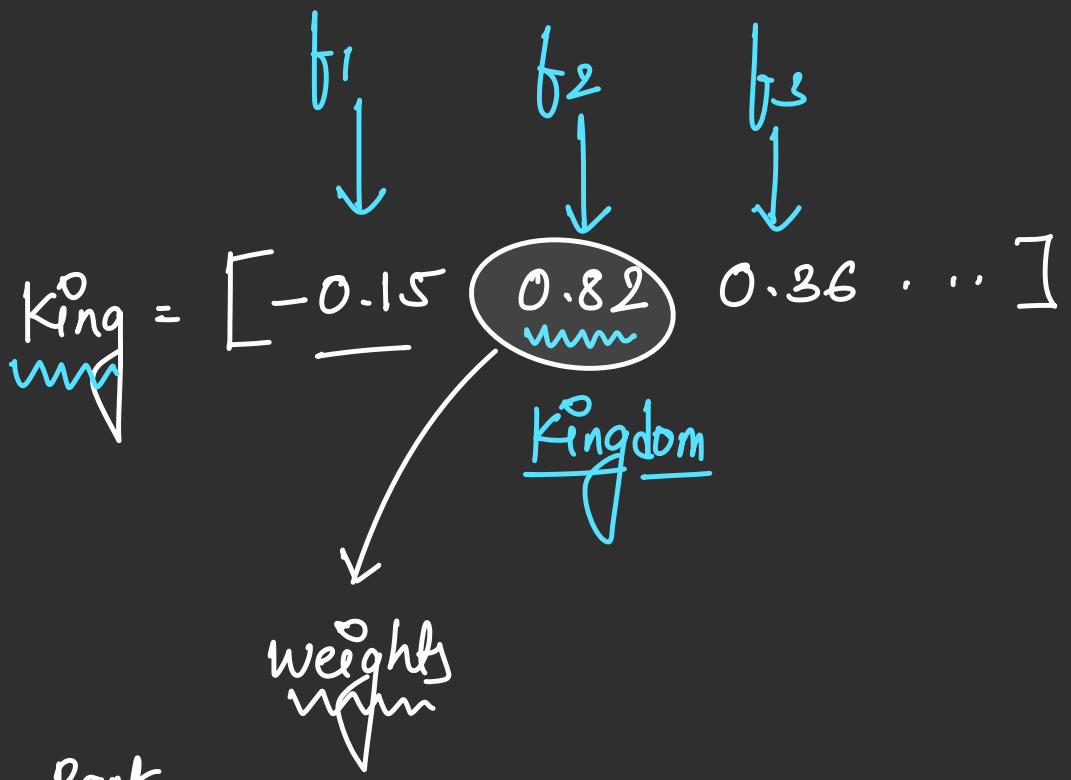


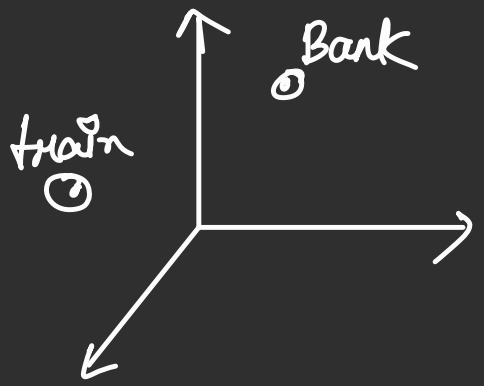


Ways to get Embeddings

1) TF-IDF
2) Word2Vec ✓✓
3) Glove
4) OpenAI Embedding ✓✓

NN





I love pizza

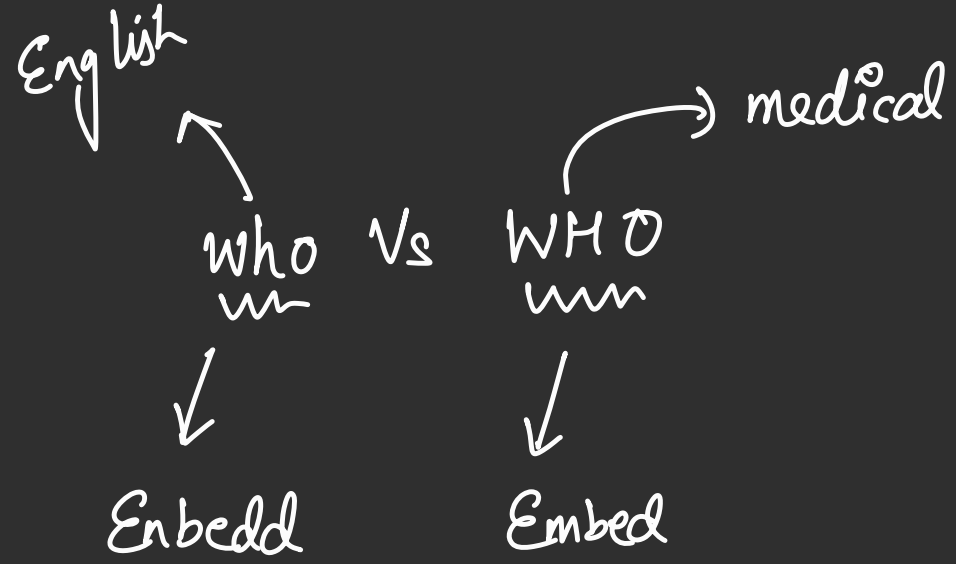
[5 23 11]

[[0.15, 0.18, 0.66],
[0.64, 0.32, 0.18]
[0.11, 0.11, 0.92]]

3x3

10,000 x 128 = 1,280000

Embedding dimension


English
who
Vs
WHO
medical
Enbedd
Embed

icecream kingdom

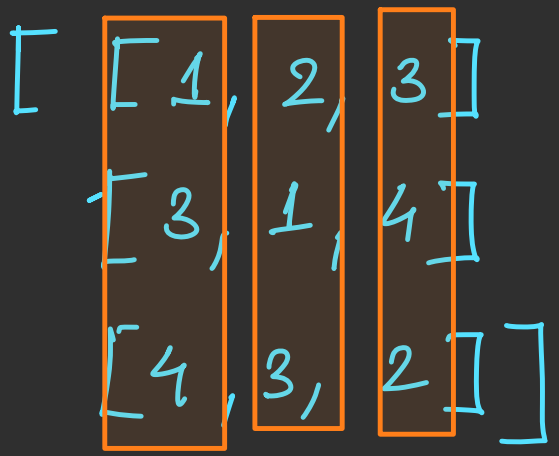
f_1 f_2 f_3 f_4 f_5 f_6 f_{128}

✓ king [0.13 0.82] = 9

✓ prince 0.23 0.78

10,000 × 128

Global Average Pooling

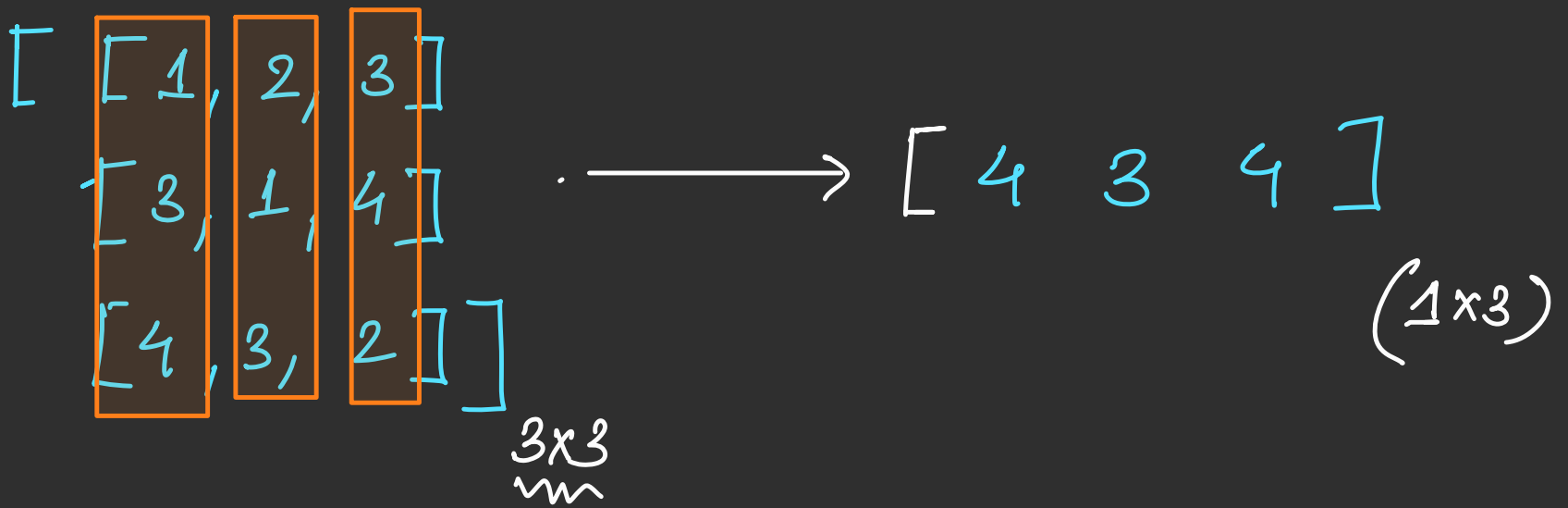


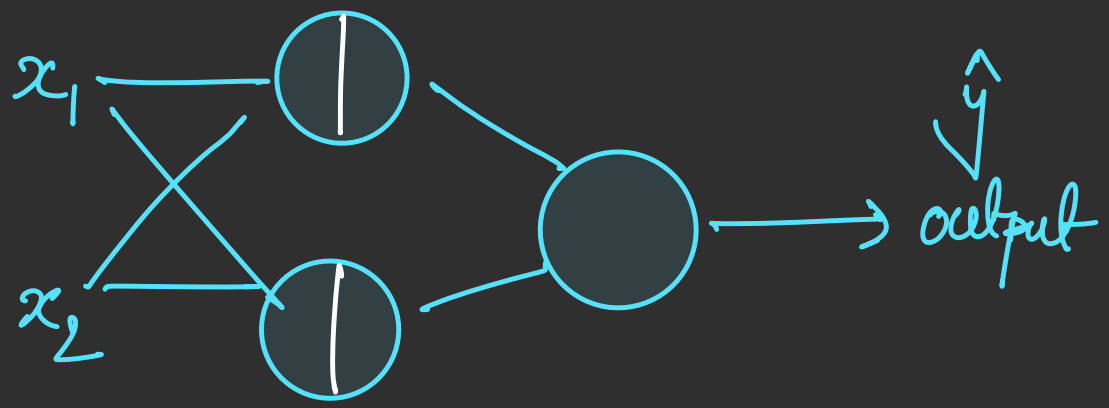
3x3

$$\rightarrow \left[\frac{(1+3+4)}{3}, \frac{(2+1+3)}{3}, \frac{(3+4+2)}{3} \right]$$

(1x3)

Global Max Pooling





$$z = w_1 x_1 + w_2 x_2 + b$$

$\hookrightarrow$ act(z)

$\hookrightarrow$ Relu / tanh / leaky

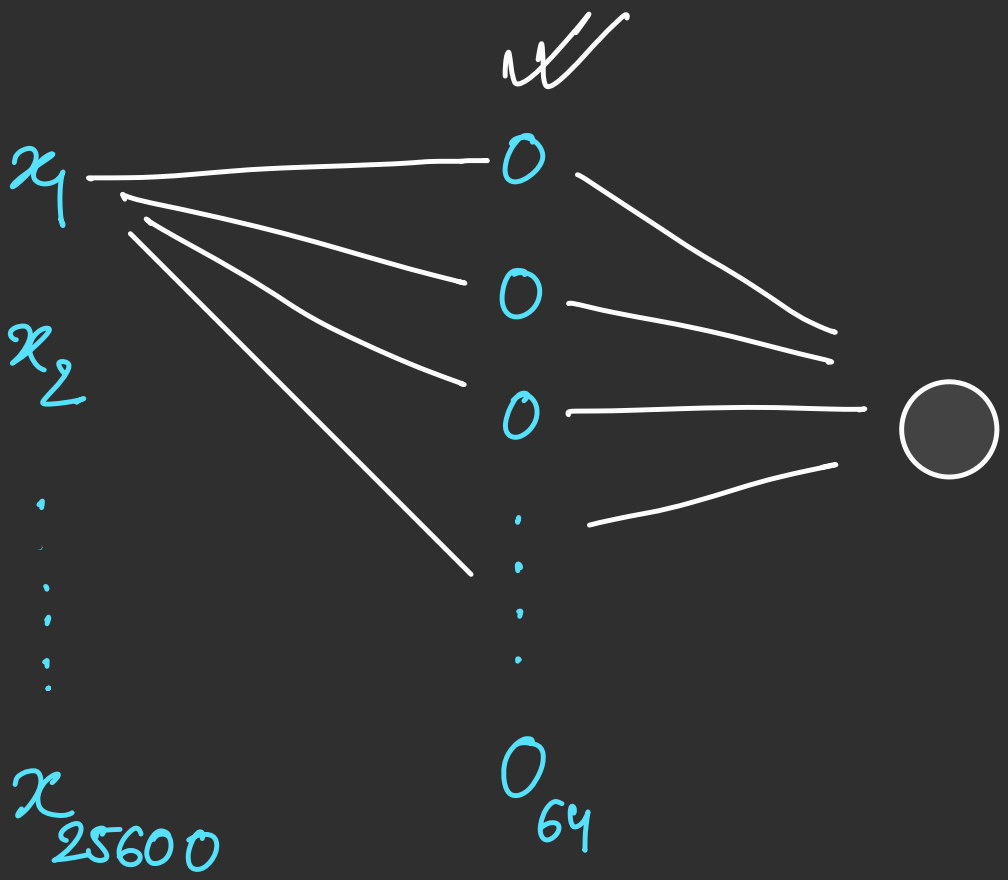
Non - linearity

embedding → Max row x max length x out dim
(10,000) (200) (128) → (Embedding dim)

(10000 x 200 x 128)

flatten → 10000 x 200 x 128 ⇒ 25600 ✓
(features)

[[3, 2, 1
1, 6 4]] → [3 21 16 4]



$123,321$

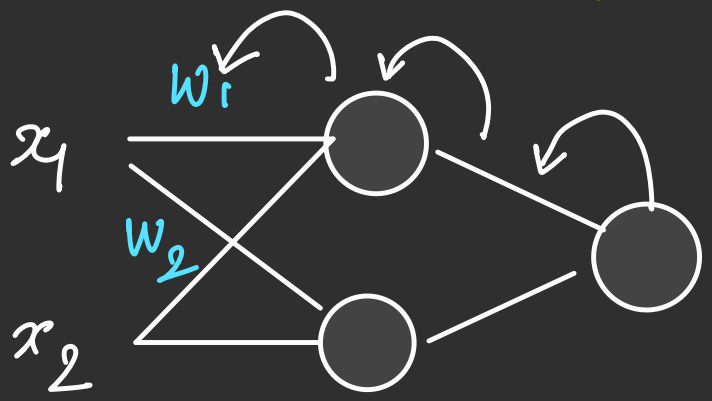
Doubts

1) Words → w_1 w_2 w_3 → Sen 1 → Take avg (w_1, w_2, w_3)

2) Using LSTM / RNN

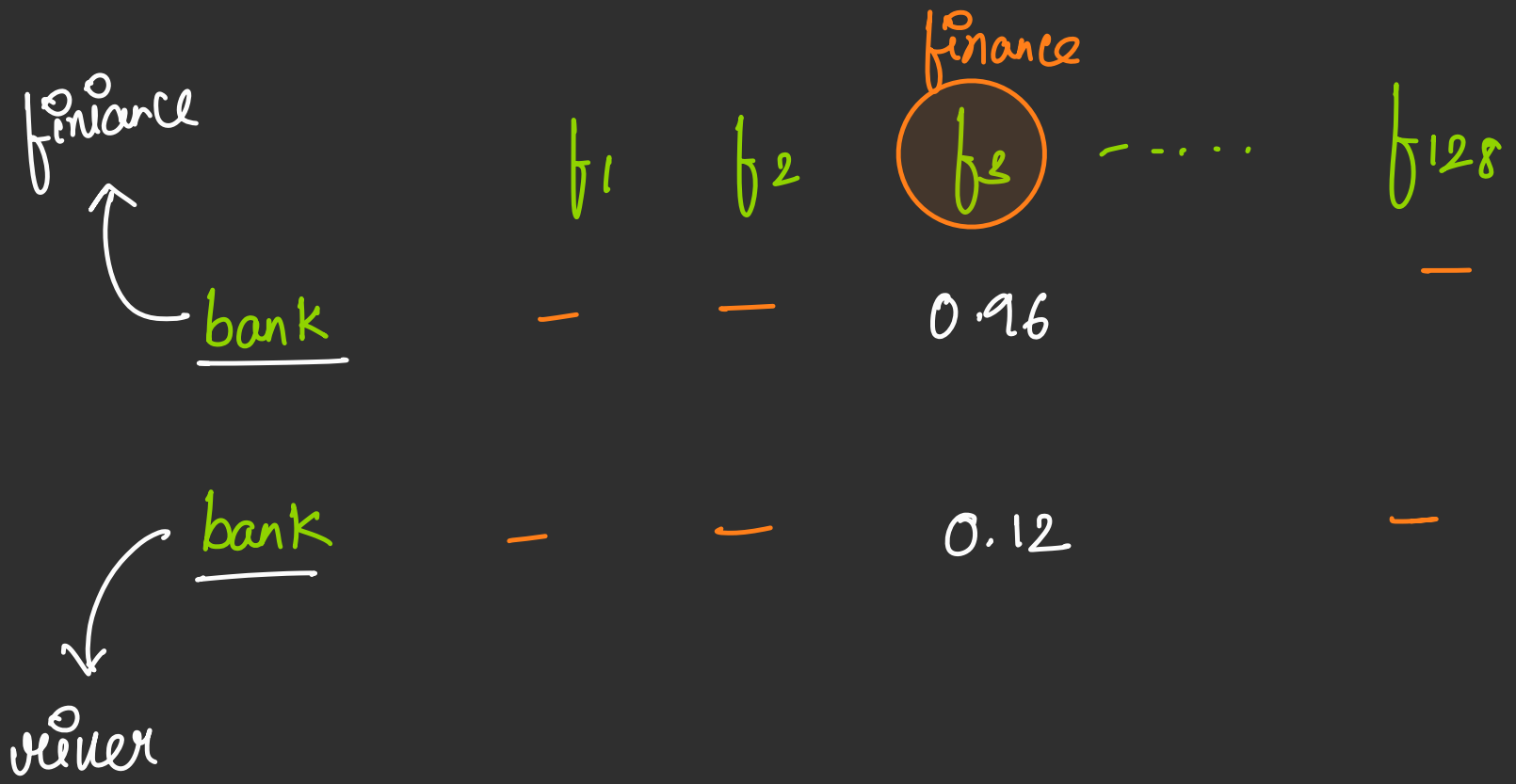
3) Transformer Architecture

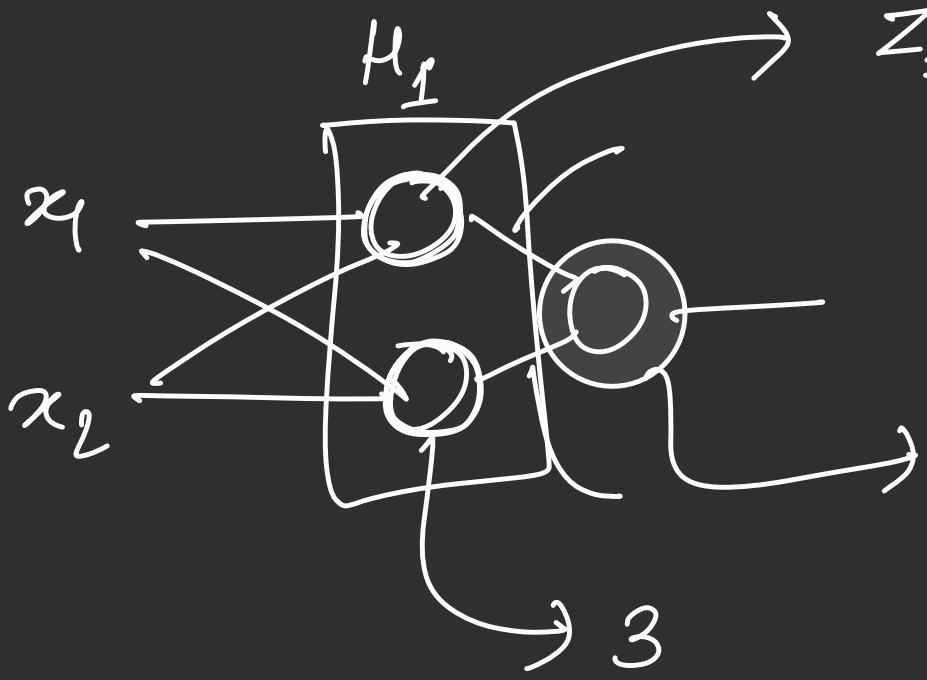
Forward prop



Back propagation (training)

Algo to train NN


finance
finance
f1
f2
f3
.....
f128
bank
0.96
bank
0.12
river



$$z_1 = w_1 x_1 + w_2 x_2 + b$$

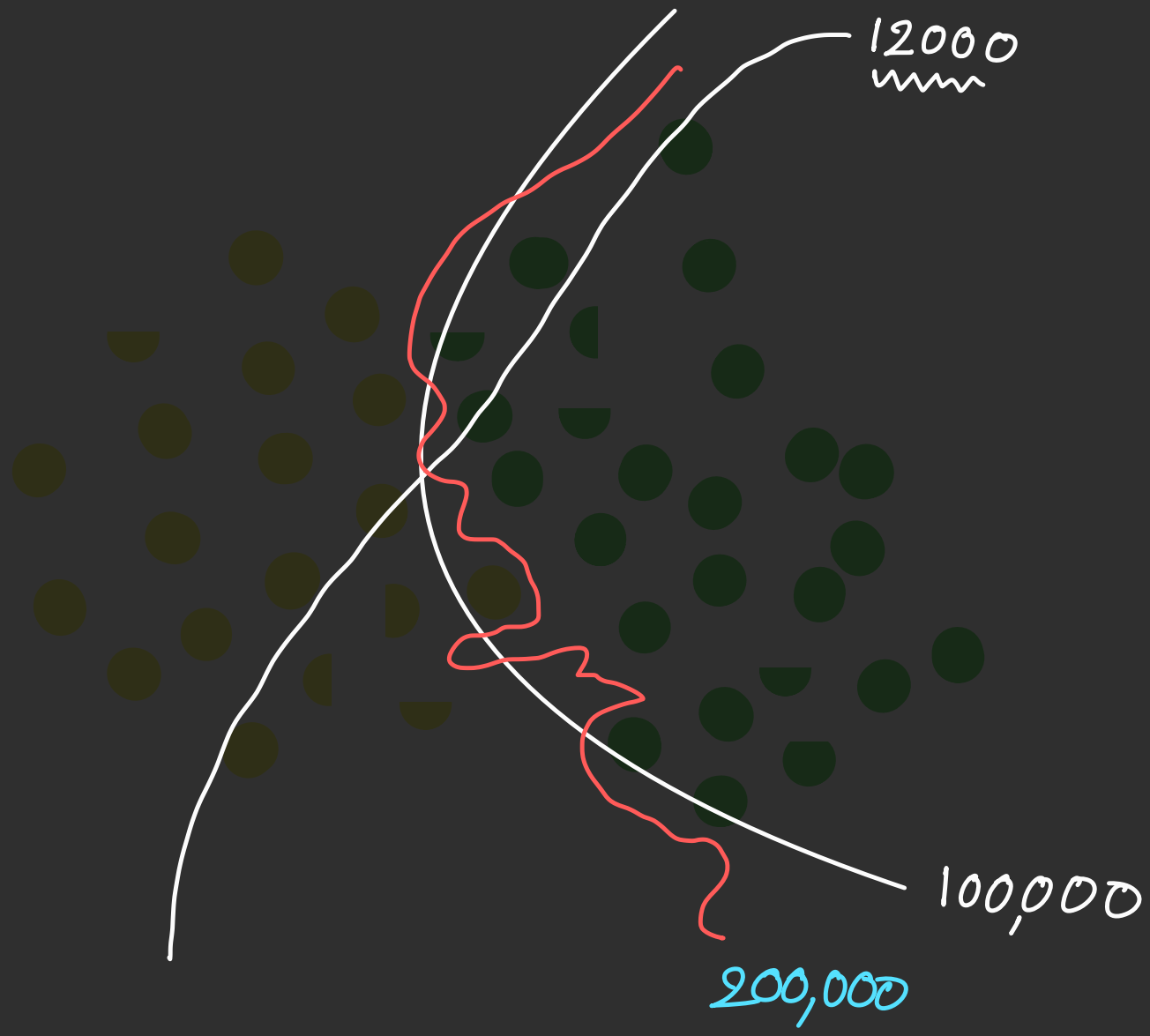
$$\Rightarrow (3)$$

$$z_2 = w_1' o_1 + w_2' o_2 \pm b$$

$$\Rightarrow$$

$$H_1 \Rightarrow 6 \quad \Rightarrow$$

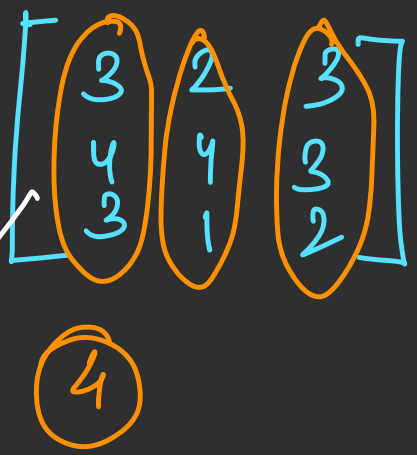
$$6 + 3 \Rightarrow 9$$


12000
100,000
200,000

$x_1 \longrightarrow [3, 2, 4]_{1 \times 3}$

$x_1' \longrightarrow [3 \quad 2 \quad 4 \quad \ldots\ldots \quad 8]_{128}$

GMP

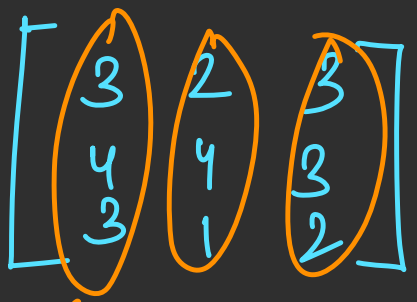


3

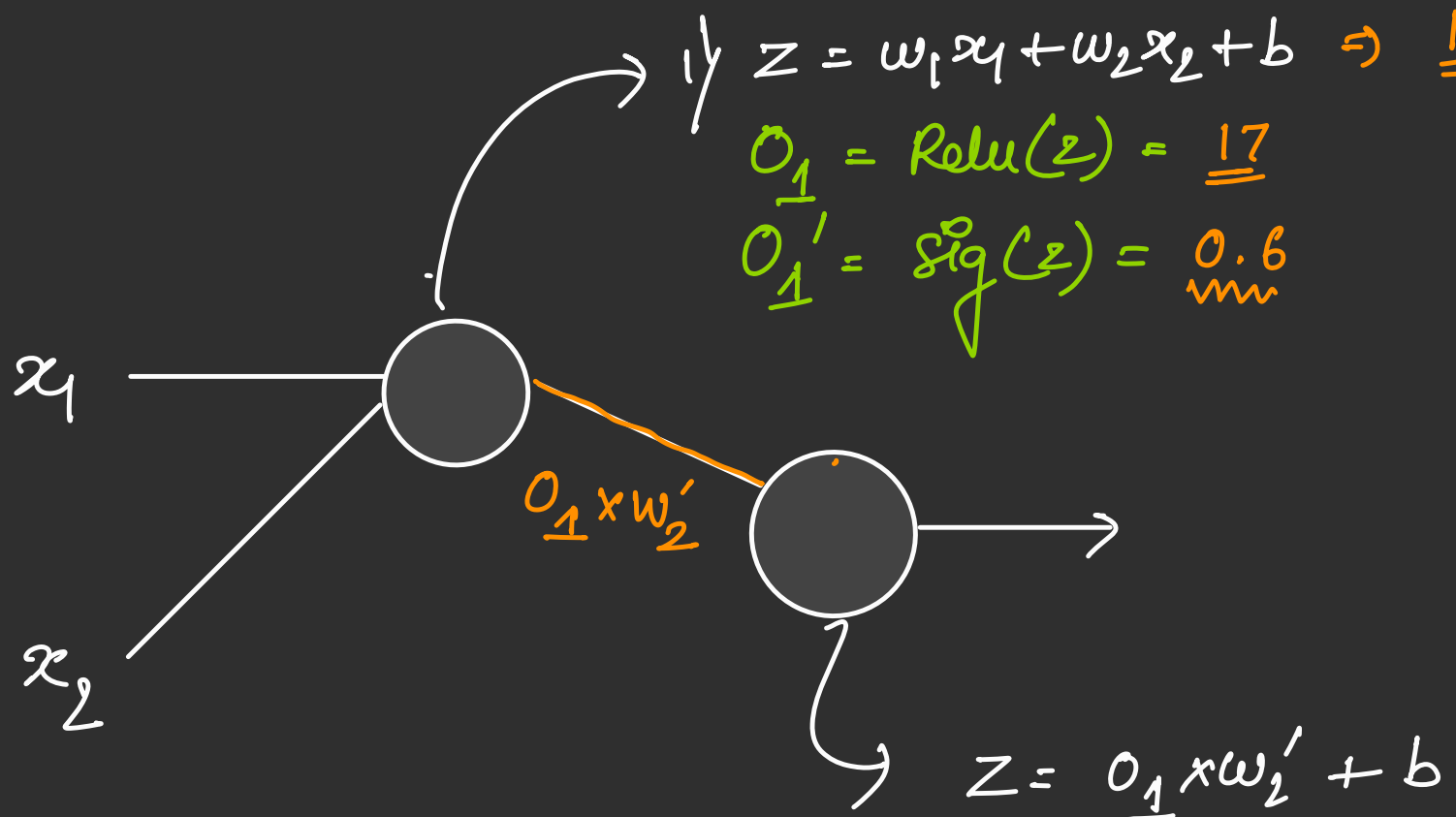
100

20

GAP



$$\frac{3+4+3}{3} = 3.33$$



"Vashing gradient"